Assignment 1 – Git & GitHub Fundamentals

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Repository: https://github.com/youssef-dev-lab/Assignment-SWE-Practice-

1. Repository Setup

I initialized a local Git repository on my Mac and used a .gitignore file to exclude environment files and ordinary log files. The rule !important.log makes important.log an exception, so it remains visible to Git and can be committed. I verified the repository state with git status.

2. README & Commit History

I created a README.md describing the project, author, technologies, example Git commands, and a link to the Git documentation. I used meaningful commit messages to record separate steps of the work. The history included commits for the ignore rules, README, tracking important.log, About documentation, and the selective-staging exercise.

```
Example commands:
git log --oneline --graph --all
git status
```

3. Selective Staging

I modified README.md and ABOUT.md. I staged ABOUT.md with git add ABOUT.md and committed that file while leaving the README.md change unstaged. The status output showed the remaining README.md modification under “Changes not staged for commit.” I later committed the README change separately.

```
Commands used:
git add ABOUT.md
git status
git commit -m "Expand About documentation"
```

4. Branch & Merge

I created the feature/about branch, added ABOUT.md, and committed the About documentation on that branch. I then switched back to main and merged feature/about. Git performed a fast-forward merge, so main and feature/about pointed to the same commit after the merge; a separate merge commit was not created.

```
Commands used:
git switch -c feature/about
git switch main
git merge feature/about
```

5. Remote Changes

I connected the local repository to the GitHub remote named origin and pushed main with git push -u origin main. A commit records changes in the local repository, while a push uploads local commits to the remote repository. Fetch downloads remote updates and remote-tracking information without integrating them into the current local branch. Pull fetches and integrates the remote changes into the current branch.

Before submission: To demonstrate the complete remote workflow, edit README.md directly on GitHub, commit that change, run git fetch origin and capture the local branch being behind origin/main, then run git pull origin main and verify that the branch is up to date. Include screenshots of these states if available.

6. Fair Use of AI

I used an AI assistant to clarify Git commands, understand the purpose of staging and commits, troubleshoot the .gitignore rules, and organize the report. I reviewed the suggested commands and ran them in Terminal, checking the results with git status, git log, and the push output. I used the explanations to understand the workflow rather than treating the commands as unexplained instructions.

7. Sources & Reflection

Sources: Git documentation (https://git-scm.com/doc) and the Assignment 1 – Git & GitHub Fundamentals instructions and report template.

The hardest concept was understanding the difference between the working tree, staging area, local commit history, and remote repository. If Git behaves unexpectedly, I would first run git status, then inspect the recent history with git log --oneline and check the remote with git remote -v.